HydroLoad-AI

An AI Solution to supply chain management



Capstone Project · Team Rhine-Riders · October 2026

By :
Prithvi Dattatreya Hegde
Nabil El Fadel

Meet the Team



Prithvi Dattatreya Hegde

Data Scientist & Statistician

M.Sc. Statistics

Experience
6 years in healthcare, academic research and business domain

Expertise
Statistical modelling, SEM, survey methodology, predictive analytics

Tools
Python · R · SQL · SPSS - Git/GitHub

Current focus
ML, NLP and Dashboard, LLMs and RAG for chatbots



Nabil El Fadel

Data Scientist & IT-Management

M.Sc. Business Informatics

Experience
Business analysis, BI and risk advisory — Vodafone, PwC and Deloitte

Expertise
BI and reporting, data-driven process optimisation, fraud analysis

Tools
SQL · Power BI · Python · Pandas · Git/GitHub

Current focus
EDA, ML, NLP and Streamlit — Data Science & AI

Agenda

01 Business Problem and Product Goal

02 Data & Method

03 Exploratory Data Analysis

04 Model and Results

05 Limitations

06 Live Demo

07 Future Work

Business Problem and Product Goal

01 CHALLENGE

Complicate Cargo Planning

Changing water levels at Kaub make it hard to know how much cargo a vessel can carry days in advance

02 FOCUS

Kaub: a key navigation gauge

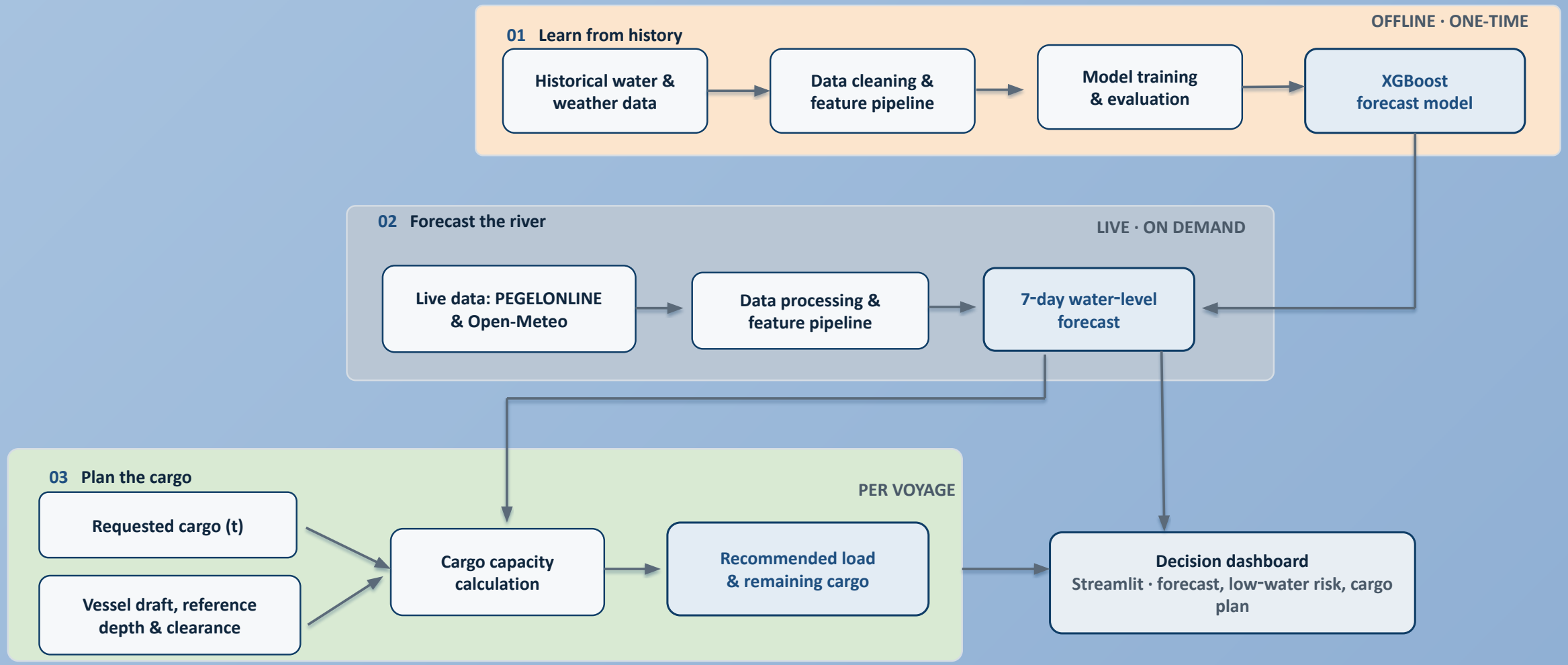
Water conditions at Kaub are operationally important for navigation through the Middle Rhine, so it is our forecasting target.

03 SOLUTION

HydroLoad-AI

It forecasts water level 7 days ahead and it integrate with the cargo capacity.

HydroLoad-AI Flow :



Data Sources and Processing

SOURCES

PEGELONLINE API

Open-Meteo

ONE REPRODUCIBLE PIPELINE

1. **Align clocks**
2. **Resolve 104 duplicate timestamps**
3. **Aggregate to days**
4. **Drop partial days**

OUTPUT

DAILY DATASET

9,740

days

Jan 2000 – Sep 2026
Along with this 13 features are considered.

Preventing Data Leakage



1 Chronological splitting only

With lag-1 autocorrelation at 0.98, a random split would leak the answer

2 Seven-day embargo between every split

The last training day's target falls seven days later inside the next block

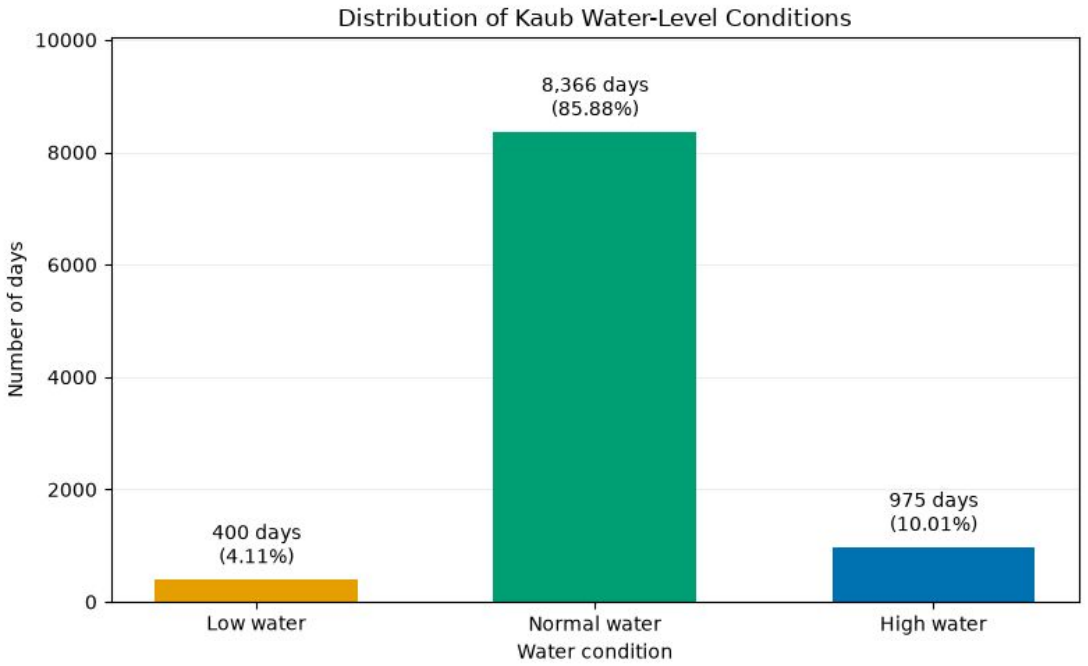
3 Backward-looking features only

- Every rolling window and lag uses data available at prediction time
- Verified manually on individual rows

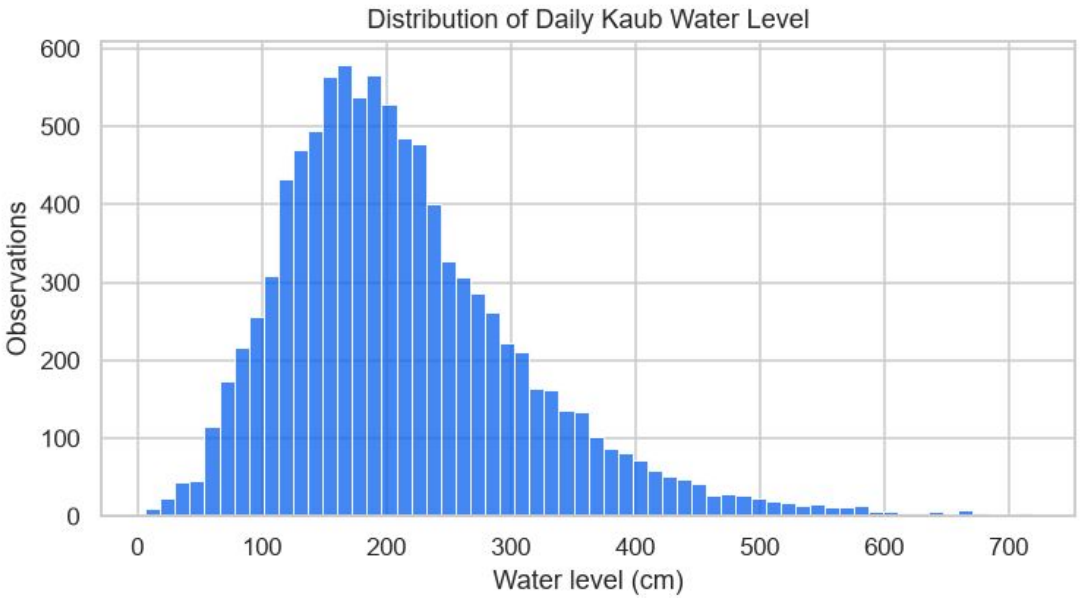
4 Test set held back

Model selection runs on validation only, so the test set stays uncontaminated

Distribution of Water Levels

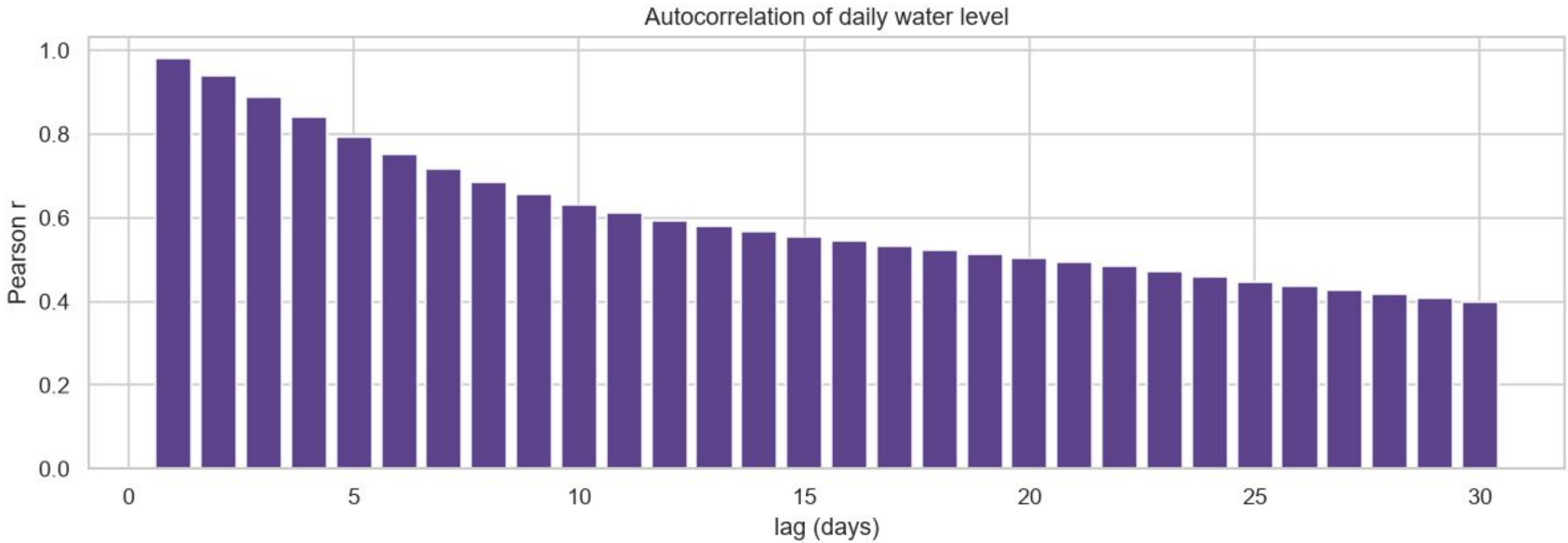


Normal water conditions dominate the data (85.88%), while low-water days account for only 4.11%.



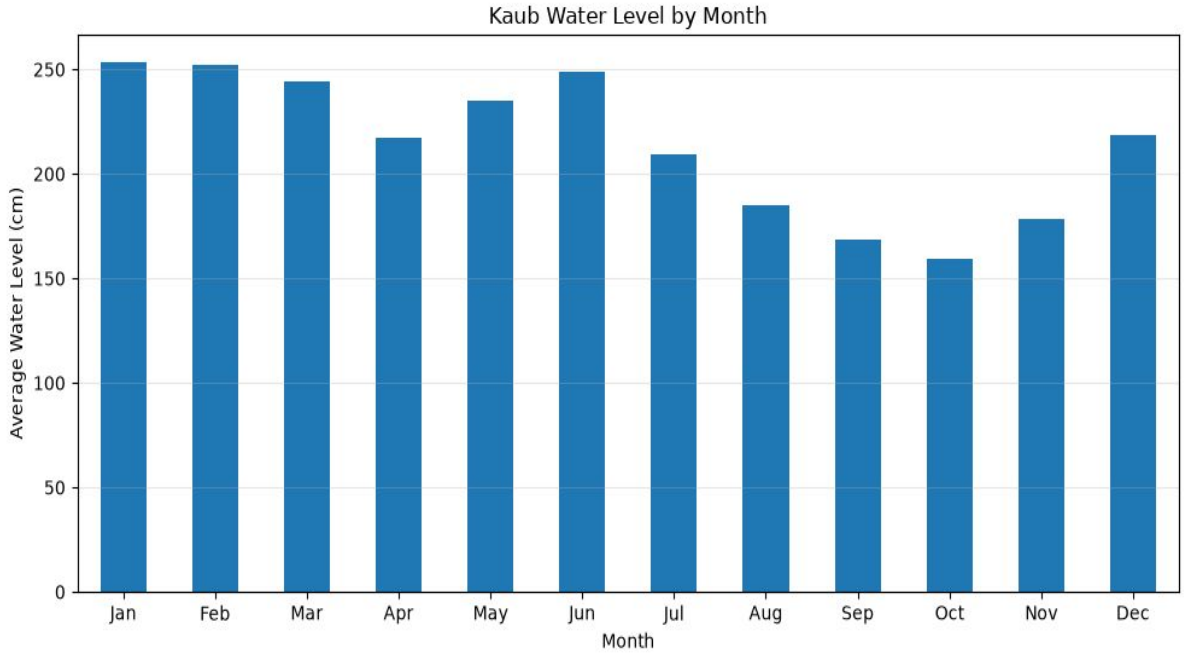
Kaub water levels are right-skewed: most observations are around 100–300 cm, with relatively few extreme high-water events.

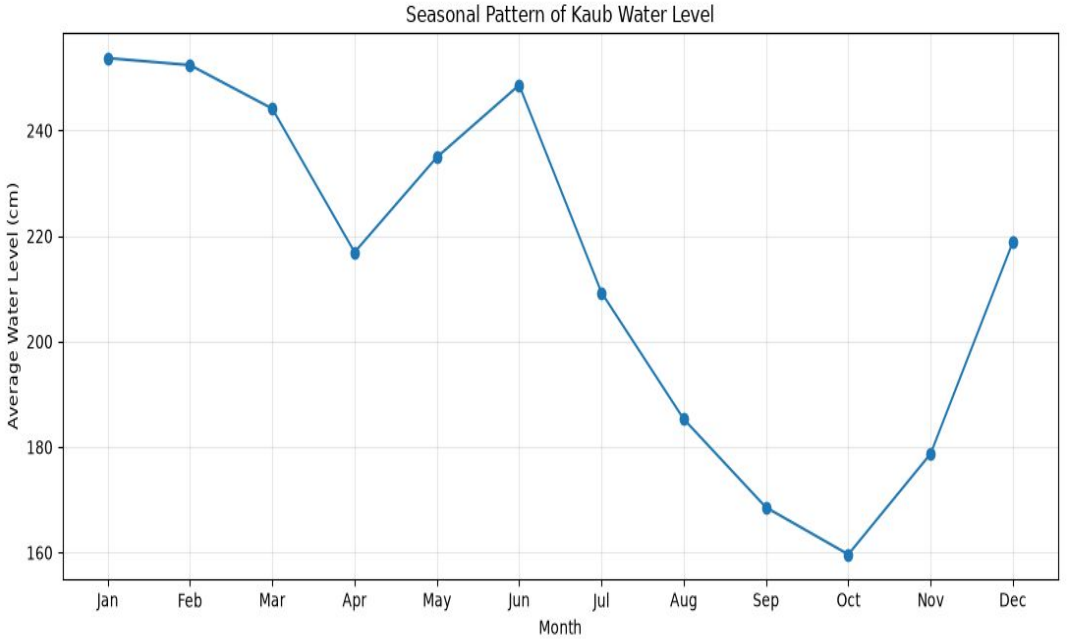
Autocorrelation of the Water Level



The high positive autocorrelations at the first several lags show strong autocorrelation, meaning the current Kaub water level is strongly related to its recent past values.

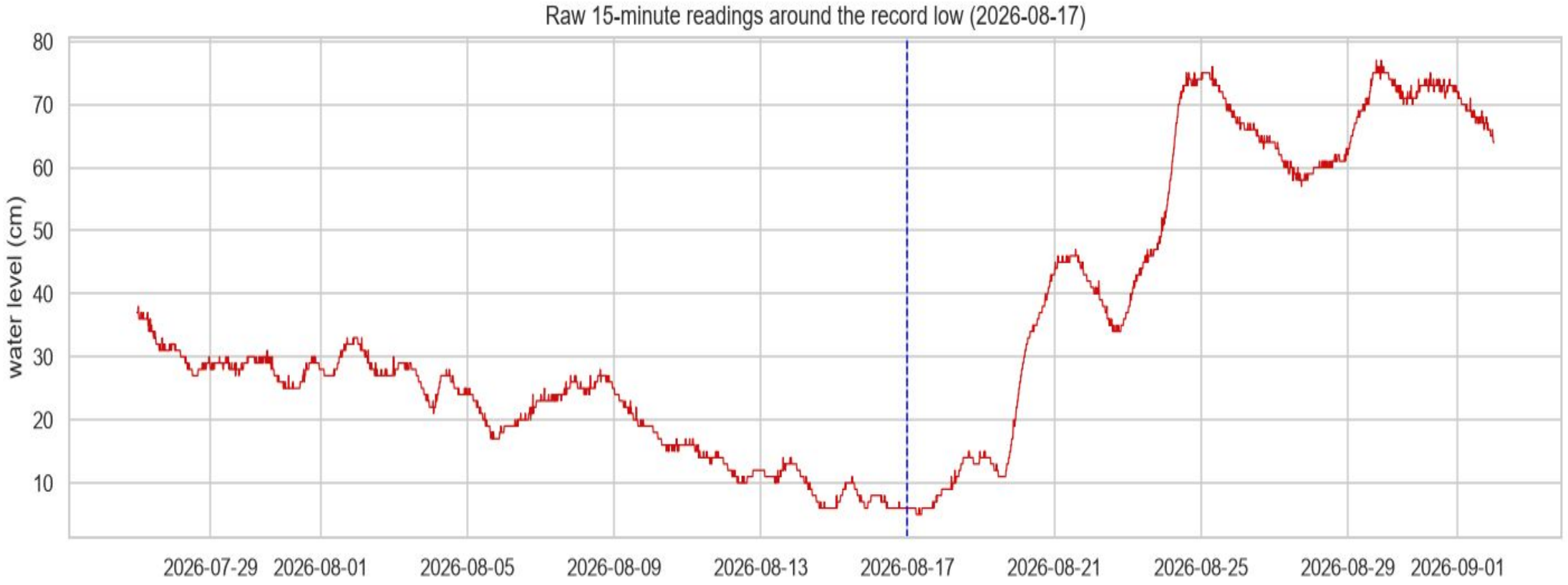
Seasonal Pattern





We have observed that there is always low water trend during the autumn across the years.

Record Low Water Level in 2026



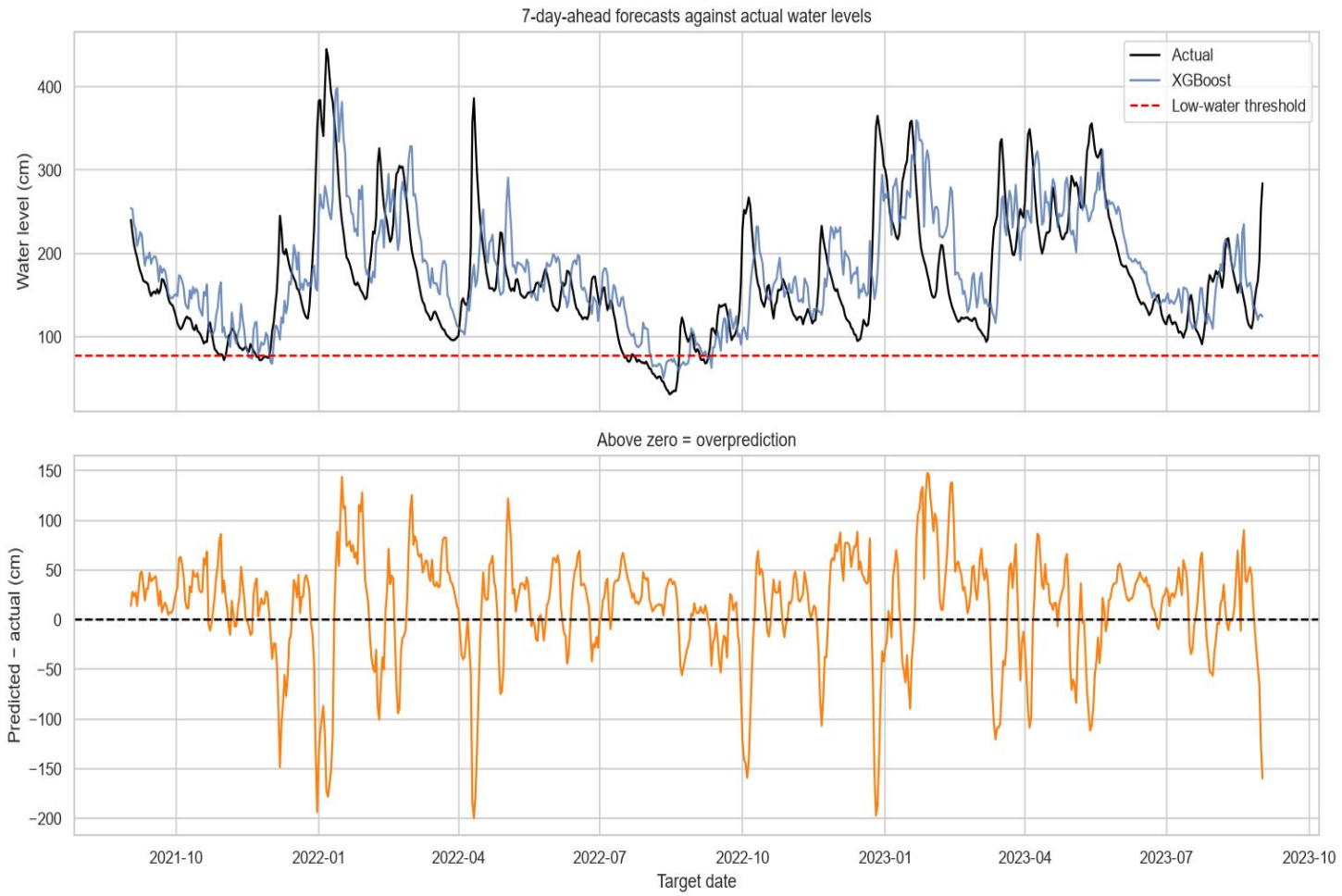
The 15 minutes data shows a record-low period around 17 August 2026, followed by a rapid rise in water level within a few days.

Model Training and Results

Approach	MAE (cm's)	RMSE (cm's)	R2	Bias
Baseline (Persistence Model)	50.83	72.46	0.583	-0.943
Univariate (Regression)	45.44	57.37	0.409	11.68
Multivariate (Regression) (Kaub+Precipitation+Temperature)	45.16	56.57	0.425	15.43
LightGBM (Tuned) (Kaub+Precipitation+Temperature)	44.55	56.8	0.421	13.65
Random Forest (Tuned) (Kaub+Precipitation+Temperature)	45.49	57.9	0.398	14.38
XGBoost (Tuned) (Kaub+Precipitation+Temperature)	**43.69**	**56.23**	**0.432**	**11.98**

XGBoost achieved the lowest validation error, offering a modest improvement over persistence, and was selected for final testing.

Model Performance



Key observations

- The model captures general water-level patterns but struggles with rapid changes.
- Overprediction during low-water periods could overestimate cargo capacity, highlighting the need to account for forecast uncertainty.

Validation window Aug 2021 – Aug 2023; XGBoost with Kaub, precipitation and temperature features.

Cargo Advice :

1

Predict the water level

Estimate the water level at Kaub seven days ahead.

2

Estimate the river depth

Use that prediction and a known reference depth to estimate how much water is available.

3

Leave space below the vessel

Subtract a clearance margin so there is space between the vessel's bottom and the riverbed.

4

Calculate how much cargo fits

Use the vessel's specifications to estimate how much cargo it can carry at that depth.

5

Suggest the load

Compare this capacity with the cargo requested and show how much can be loaded and how much remains.

Limitations and improve the accuracy



Limited extreme-water data

Low-water and high-water conditions account for only 4 % and 10 % of the data, which limits the model’s ability to learn these patterns.



More weight on low water

Giving low-water days more weight in training can improve accuracy where it matters most.



Additional weather inputs

Incorporating precipitation forecasts could help the model anticipate changes in water levels.



Changing river conditions

Variables such as river discharge and flow velocity could help capture river dynamics.

Future Plans: Scaling the Solution

From a Kaub prototype to a Rhine-wide decision tool for freight operators

Geographic Expansion

Apply the same pipeline to further bottleneck gauges such as **Maxau** and **Koblenz** — upstream gauges also give the Kaub forecast real lead time.

Advanced Forecasting

Add weather forecasts and Alpine snowmelt to extend the horizon from **7 to 14 days**. A rainfall forecast alone lifted day-7 skill from **10 % to 30 %** in our test.

Cost Optimization

Turn tonnes into euros: forecast **low-water surcharges**, charter and lorry costs, and recommend the cheapest safe mix of ship, rail and road.

Smart Dashboard

Deploy the app as a web service with an **AI assistant** that answers “How much can we load on Friday?” Plus automatic low-water alerts.

Live Demo

HydroLoad-AI

RHINE CARGO CAPACITY ADVISOR

Data source

- (•) Live forecast
- ( ) Historical forecast

Low-water threshold (cm)

77 − +

Forecast horizon: 7 days

Cargo planning

Requested cargo (t)

2000.00 − +

› Vessel and depth assumptions

Deploy ⋮

Rhine Water-Level Forecasting and Cargo Decision Support



Fetch the latest water and weather observations and generate an updated forecast.

Update live forecast

Last successful fetch: 08 October 2026, 19:01 CEST. Click Update live forecast to fetch again.

Behind the forecast

Thank You – Questions?

We believe in networking! Please scan to connect…



Prithvi Dattatreya Hegde

M.Sc. Statistics

LinkedIn

linkedin.com/in/prithvihegde19940523



Nabil El Fadel

M.Sc. Business Informatics

LinkedIn

linkedin.com/in/nabil-el-fadel-b30018267

Project repository

HydroLoad-AI on GitHub

GitHub

github.com/PrithviHe/Rhine-cargo-advisor